Cyclic dominance of a Potts–RPS model on complex networks

Full test and results report

Reproducible run 2026-07-01 · 16/16 steps OK · all figures + data regenerated byte-for-byte identical (deterministic seeds)

Model: q=3 Potts spins (Rock/Paper/Scissors) on ER and BA graphs; payoff $P = I + \varepsilon\,\mathrm{skew}$ (cyclic-dominance strength ε); Glauber update at T=0.65; order parameter $m_\psi = |\langle r + p\,e^{i2\pi/3} + s\,e^{i4\pi/3}\rangle_t|$ ($m_\psi \to 1$ ordered/consensus, $\to 0$ cycling). Three descriptions: **MC** (agent-level ground truth), **HMF** (homogeneous mean field), **DMF** (degree-based mean field). ε_c quoted at the m_ψ=0.5 crossing.

1. Run manifest and reproducibility

Every step below was produced in one clean run of `./run_all.sh` (clean-rebuild C++ engine → validation test → all simulations). Per-step logs live in `logs/`; the machine-readable manifest is `logs/manifest.csv`.

step	status	wall (s)
test_validate_engine	OK	4
mean_field_hmf	OK	1
mean_field_sweep	OK	6
mean_field_compare_suite_ER	OK	6
mean_field_compare_suite_BA	OK	6
monte_carlo_mc	OK	3
monte_carlo_compare	OK	79
phase_diagram_ER	OK	6
phase_diagram_BA	OK	6
dynamics_ternary	OK	2
dynamics_fss	OK	3
dynamics_stability	OK	0
zealots_experiment	OK	3
zealots_experiment_hubs	OK	4
zealots_experiment_mixed	OK	2
defects_experiment	OK	8
total	**16/16 OK**	**139**

Table 1: All simulation steps, status and wall-clock time (16-core host, C++20 engine `-O3 -march=native`). After the run `git status` reports zero changes to any tracked figure/CSV: fully deterministic.

2. Engine validation (C++ vs pure-Python MC)

Same graph through both engines; independent RNG streams, so agreement is in phase not last digit (log: `test_validate_engine.log`).

ε	Python m_ψ	C++ m_ψ	regime / verdict
0.20	0.999	0.999	agree (ordered)
0.50	0.991	0.992	agree (ordered)
0.70	0.004	0.007	agree (cycling)
0.90	0.004	0.003	agree (cycling)

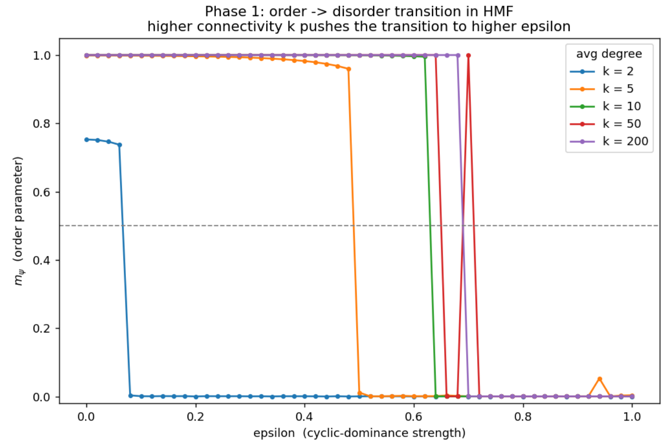
Table 2: Engine agrees in every regime; measured speedup ≈ **20**× (Python 3.47 s vs C++ 0.17 s for the 4-point sweep).

3. Mean field and Monte Carlo (recreation core)

3.1 HMF ε-sweep: connectivity stabilises order

$\langle k \rangle$	2	5	10	50	200
ε_c	0.08	0.50	0.64	0.66	0.70

ε_c rises monotonically with mean degree $\langle k \rangle$: denser networks sustain order against stronger cyclic pressure.



3.2 MC vs HMF vs DMF: RMSE against ground truth ($\langle k \rangle$=10)

graph	RMSE(HMF)	RMSE(DMF)	DMF gain	ε_c(MC)	ε_c(MF)
ER	0.338	0.334	**0.004**	0.52	0.64
BA	0.336	0.328	**0.008**	0.52	0.64

Table 3: DMF beats HMF on both graphs; the DMF advantage is $\sim$2$\times$ larger on heterogeneous BA. Both mean fields overestimate the ordered phase (MC turns over at $\varepsilon \approx 0.52$ vs ≈ 0.64).

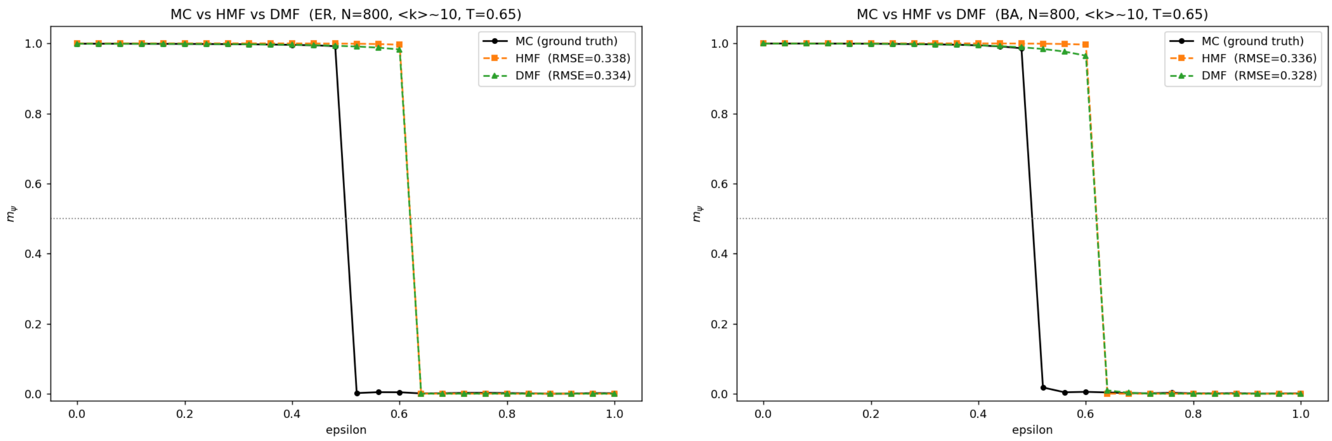
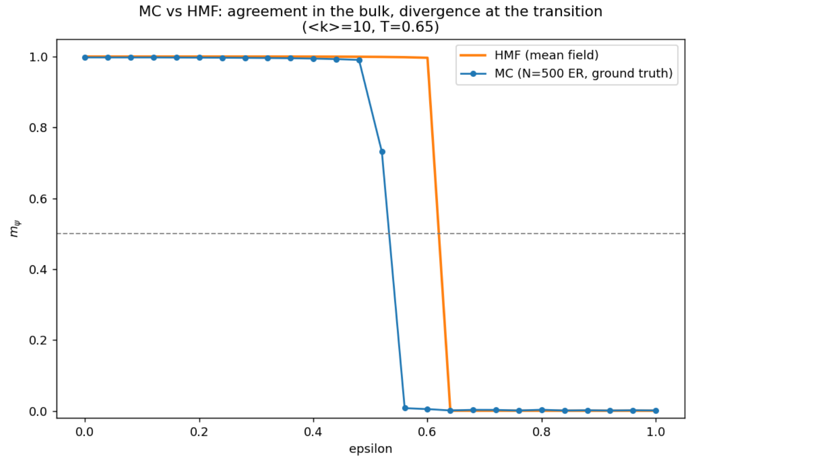


Figure 1: Comparison suite, ER (left) and BA (right).

3.3 Direct MC vs HMF overlay (ER, $\langle k \rangle$=10)

model	ε_c
MC (ground truth)	0.56
HMF	0.64

MC transition precedes HMF: the mean field overestimates the ordered region, as expected on a finite network.



4. Phase diagram: ($\langle k \rangle \times \varepsilon$) MC sweep

520 simulations per graph (20 degrees × 26 ε), N=800, fanned across 16 cores in ∼6 s each.

$\langle k \rangle$	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36
ε_c (ER)	0.00	0.24	0.40	0.48	0.52	0.56	0.60	0.60	0.60	0.64	0.64	0.68	0.68	0.68	0.68	0.68	0.68	0.68
ε_c (BA)	0.00	0.24	0.40	0.48	0.52	0.56	0.60	0.60	0.60	0.64	0.64	0.68	0.68	0.68	0.68	0.68	0.72	0.72

Table 4: Transition ε_c vs mean degree for ER and BA. The order–cycling boundary bends upward with $\langle k \rangle$; ER and BA are nearly identical ⇒ for MC dynamics average degree dominates over the shape of $P(k)$.

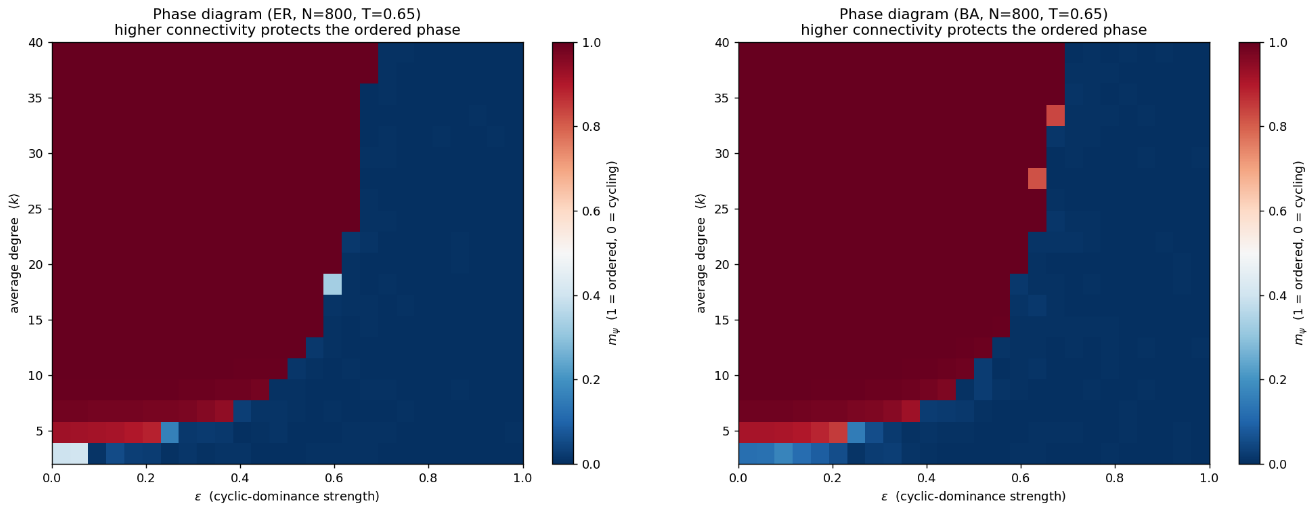


Figure 2: MC phase diagrams: ER (left), BA (right).

5. Dynamics and finite-size scaling

N	200	500	1000	2000
ε_c	0.570	0.530	0.510	0.510
max|slope|	33	49	49	44

Table 5: Finite-size scaling (ER, $\langle k \rangle$=10): the transition sharpens and ε_c converges toward ≈0.51 as N grows 200→2000.

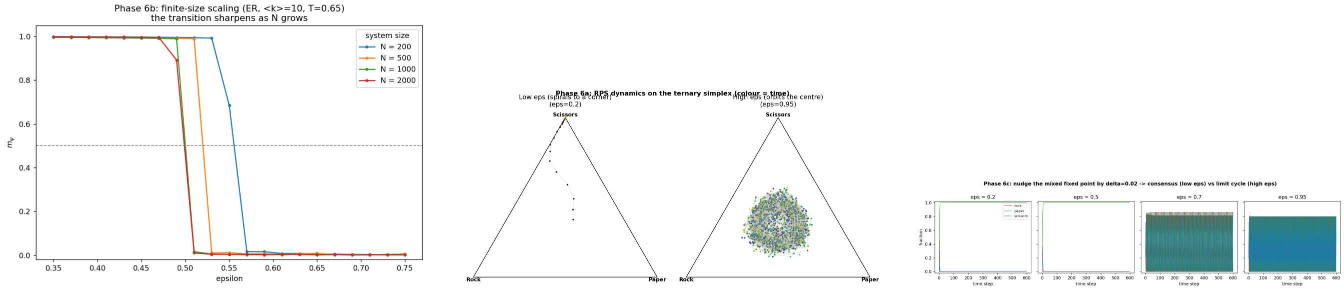


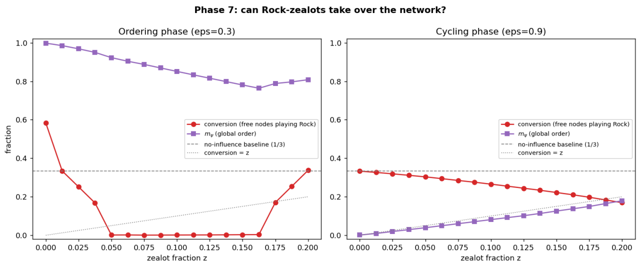
Figure 3: Left: FSS. Middle: ternary phase portrait (corner consensus below ε_c, limit cycle above). Right: fixed-point stability.

6. Perturbation experiments (new results)

6.1 Single Rock-zealot faction (ER, $\langle k \rangle$=10, 12 seeds)

z	conv_{ord}	$m_{\psi\,\text{ord}}$	conv_{cyc}	$m_{\psi\,\text{cyc}}$
0.00	0.58	1.00	0.33	0.00
0.05	0.00	0.92	0.30	0.04
0.10	0.00	0.85	0.27	0.08
0.15	0.00	0.78	0.22	0.13
0.20	0.34	0.81	0.17	0.18

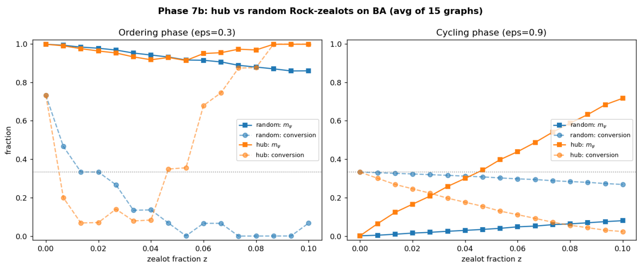
conv = fraction of *free* nodes playing Rock. Ordering phase: Rock-conversion collapses to ∼0 by z≈0.05 — the free network flips to **Paper** (Rock's predator). Cycling phase: only weak induced order, strategy cannot be pinned.



6.2 Hub vs random placement (BA, 15 seeds)

z	$m_{\psi\,\text{cyc}}^{\text{rand}}$	$m_{\psi\,\text{cyc}}^{\text{hub}}$	$m_{\psi\,\text{ord}}^{\text{rand}}$	$m_{\psi\,\text{ord}}^{\text{hub}}$
0.00	0.00	0.00	1.00	1.00
0.03	0.02	0.21	0.97	0.95
0.05	0.04	0.40	0.92	0.91
0.08	0.06	0.59	0.88	0.97
0.10	0.08	0.72	0.86	1.00

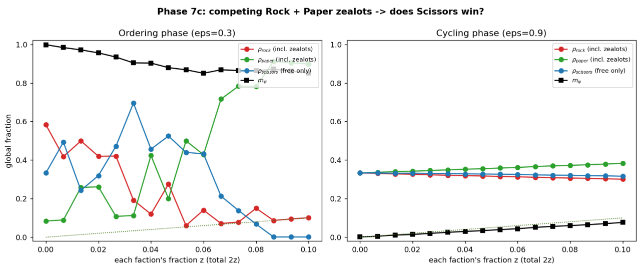
Cycling phase at z=0.10: hub m_ψ= 0.72 vs random = 0.08 — a **8.9× amplification**. Hubs control *whether* the network orders, not *what* it orders on.



6.3 Competing Rock+Paper factions (ER, 12 seeds)

z (each)	ρ_{rock}	ρ_{paper}	ρ_{sciss} (ord.)
0.00	0.58	0.08	0.33
0.03	0.42	0.11	0.47
0.05	0.06	0.50	0.44
0.08	0.15	0.78	0.07
0.10	0.10	0.90	0.00

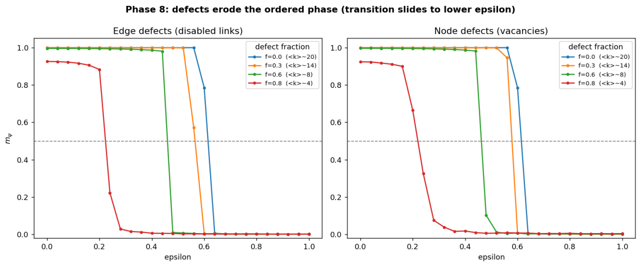
With equal Rock+Paper zealots, the ordering-phase free population goes to **Paper** (ρ≈0.90), *not* Scissors: Paper is reinforced both by its own zealots and by the Rock-zealots it preys on. Cycling phase stays robust (m_ψ≈0.08).



6.4 Network defects: edge vs node quenching (ER, 6 seeds)

f	ε_c(edge)	$\langle k \rangle$(edge)	ε_c(node)	$\langle k \rangle$(node)
0.00	0.64	20.0	0.64	20.0
0.30	0.60	14.0	0.60	14.1
0.60	0.48	8.0	0.48	8.0
0.80	0.24	4.1	0.24	4.0

Removing edges or nodes slides ε_c down ($0.64 \to 0.24$). Edge and node defects **coincide** once matched by the resulting $\langle k \rangle$ $\Rightarrow$ order-stability depends on effective $\langle k \rangle$ only.



7. Findings summary

1. **Connectivity stabilises order:** ε_c rises with $\langle k \rangle$ in HMF ($0.08 \to 0.70$) and in the MC phase diagram ($0 \to 0.72$).
2. **DMF > HMF:** lower RMSE vs MC on both graphs, advantage $\sim 2\times$ larger on heterogeneous BA. Both mean fields overestimate the ordered phase (MC $\varepsilon_c \approx 0.52$ vs MF ≈ 0.64).
3. **Average degree, not $P(k)$:** ER and BA phase boundaries nearly coincide.
4. **Genuine transition:** FSS sharpens and ε_c converges to ≈ 0.51; phase portrait shows corner consensus vs limit cycle.
5. **Zealots back-fire:** a Rock minority flips the free network to Paper (its predator), not Rock — naive minority takeover fails in a cyclic system.
6. **Hub leverage $\sim 9\times$:** hub-placed zealots amplify induced order vs random, but set *whether*, not *what*, the network orders on.
7. **Predator wins under competition:** equal Rock+Paper zealots drive the free population to Paper ($\rho \approx 0.90$), not Scissors.
8. **Defects $\equiv$ effective $\langle k \rangle$:** edge and node damage are equivalent once matched by resulting mean degree; ε_c slides $0.64 \to 0.24$.

Generated by `build_report.py` from the regenerated CSVs and `logs/manifest.csv`. 23 artifacts (12 figures + 11 data tables). Full narrative in `FINDINGS.md`; run details in `RUN_REPORT.md`.